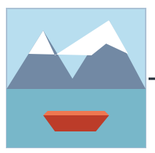
(a) Quality-aware multimodal pretraining — fused-z I–I

Image I



224 × 224 × 3

SafeAug ×2 + Clean

Shared visual encoders

CLIP ViT-L/14 → Wg: 768→256 → g (256-D)

ResNet-50 layer3 → R: 1024→256, L: 256→256 → P_0 (196 × 256)

Fl = R(layer3); L and LF are independent

Fusion: Q = g; K,V = LF(Fl)
Cross-attention → z (256-D)

Fusion used by I–I and image–text relations.

Image–Image SimSiam (symmetric)

z1 → PI → h1 → DI → p1 vs sg(h2)

z2 → PI → h2 → DI → p2 vs sg(h1)

fused z · shared PI / DI

Image–text relations

z → PI → AG → DI ↔ text targets

Prompt: T* → U → PT → AT → DT

Answer: T* → U → PT → DT

T*: frozen text tower + trainable contexts

Four SimSiam relations

I–I · Image–Prompt · Image–Answer · Prompt–Answer

Positive local / phrase supervision retained; negative push disabled.

Clean quality supervision — two independent paths

g → PQ → qg

P_0 + WNSS(n) → QL + pool → ql

q = norm[(1−η)qg + ηql]
feature mixture (not g + P_0)

Component anchors + ordinal
Artifact / Aesthetics

Separate Clean MOS supervision

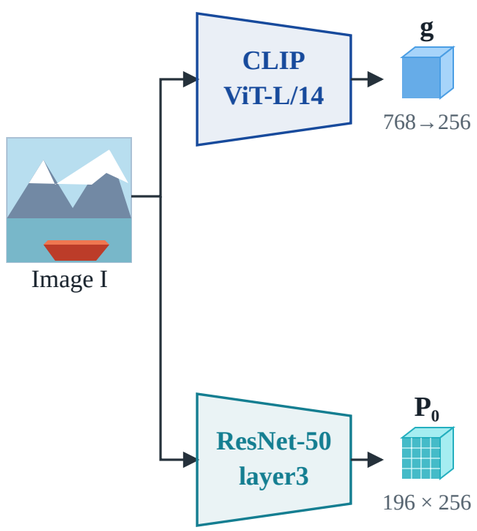
Artifact/Aesthetics heads: [qg, ql, |qg−ql|]

Quality / Align heads supervise Clean score outputs.

Transfer encoders, PQ→QA, PI/PT, adapters and local quality; reset score heads.

(b) Downstream quality and alignment assessment

Visual encoders (fine-tuned)



ResNet: 1024→256 → Fl

Local L: 256→256 → P_0

Repeated ports denote the same tensor.

Quality branch Q

Global: g → QA (from PQ) 256→512→512→256 → qg → α·sg(||g||)·norm(qg) → (+ g) → Hg + sigmoid → Qg

Local: P_0 (196 × 256) → (+ s_i) → QL + norm → o_i → Gate MLP([o_i,n_i]) softmax → w_i → ql = norm Σ_i $w_i o_i$ (inputs: w_i, o_i) → Hl + sigmoid → Ql

Q = (1−β)Qg + βQl → Quality Q

Image I → NSS-like statistics → n_i (6-D) → WNSS MLP: 6→256 → s_i (256-D) reuse n_i in gate

Alignment branch A

g → PI → AG → v → [v,u,|v−u|] → Global MLP + sigmoid → Ag

P_0 → PI → AP → p_i → p_i,u → XI(p_i)·XT(u) softmax → a_i → e = Σ_i $a_i p_i$ pool original p_i → [e,u,|e−u|] MLP + sigmoid → Al

shared PI

γ = sigmoid G([v,u,Ag,Al]) (inputs: Ag, Al, γ)

A = γAg + (1−γ)Al
Text–image alignment

Prompt + 4 contexts: T* frozen → U → PT → AT → u

Named ports v,u,p_i,Ag,Al avoid crossed wires; γ is sample-wise.

Solid arrows: forward · Purple dashed: transfer/shared weights · sg: stop-gradient · T*: frozen text tower, trainable contexts

OFF: quality text CE, negative push. ON: Soft-SROCC. PI/PT BN statistics frozen downstream; visual weights fine-tuned.